VALIDATION REPORT
INSTALLATION, OPERATIONAL AND PERFORMANCE QUALIFICATION
JR VALIDATED ENVIRONMENT — CORE

JR Validated Environment — Version 1.0

Version History

Version	Date	Author	Description
1.0	2026-03-12	Joep Rous	Initial release — v1.0 IQ/OQ/PQ execution report.

Approval Signatures

Role	Name	Signature	Date
Validation Manager	Joep Rous	JR	2026-03-12
Reviewer	Joep Rous	JR	2026-03-12

1. Purpose

This report documents the execution and outcome of the Installation Qualification (IQ), Operational Qualification (OQ), and Performance Qualification (PQ) for the JR Validated Environment v1.0. It provides objective evidence that the core environment has been installed correctly, functions as designed, and performs correctly under realistic use conditions.

This report covers the core environment only. Community scripts are qualified separately under JR-VP-002 / JR-OQ-001.

2. Reference Documents

Document ID	Title	Location

JR-VP-001 v1.0	Validation Plan — IQ/OQ/PQ, JR Validated Environment Core	docs/IQ_validation_plan.docx
IQ Evidence	IQ Validation Evidence (admin_validate output)	docs/IQ_validation_20260311_205146.txt
JR-VP-002 v1.0	Validation Plan — OQ, Community Script Suite	docs/oq_validation_plan.docx
JR-OQ-001 v1.0	Validation Report — OQ, Community Script Suite	docs/OQ_validation_report.docx
—	Administrator Manual	docs/admin_manual.docx
—	Software change history	CHANGELOG.md

3. Test Environment

Field	Value
Hardware	MacBook Air, Apple M3, 16 GB
macOS version	macOS Sequoia 15.7.3
R version	R 4.5.0
Python version	Python 3.11.9
JR env version	v1.0 (git tag: v1.0)
Tester	J. Rous
Test date	2026-02-12

4. Installation Qualification (IQ) Results

The IQ was executed using the admin_validate script. The evidence file generated by admin_validate is attached as Appendix A.

Step	Action	Expected Result	Actual Result
1	Run ./admin/admin_install_R from project root	All R packages installed at pinned versions in ~/.renv/[PROJECT_ID]/library/. renv.lock written. Verification passes.	PASS
2	Run ./admin/admin_install_Python from project root	All Python packages installed at pinned versions in ~/.venvs/[PROJECT_ID]/. Verification passes.	PASS
3	Run ./admin/admin_validate from project root	VALIDATION RESULT: PASSED. Evidence file written. R and Python versions confirmed. All packages at pinned versions.	PASS

5. Operational Qualification (OQ) Results

ID	Description	Steps	Expected Result	Result	Findings / Notes
OQ-01	R environment rebuild from scratch	1. Delete ~/.renv/[PROJECT_ID] 2. Run admin_install_R 3. Verify output	Packages installed at pinned versions. Verification passes.	PASS	3 findings — see Section 7. All resolved prior to OQ completion.
OQ-02	Python environment rebuild from scratch	1. Delete ~/.venvs/[PROJECT_ID] 2. Run admin_install_Python 3. Verify output	Packages installed at pinned versions. Verification passes.	PASS	None.
OQ-03	Run R hello-world script	1. Run: jrrun jrhello.R "Hello" 2. Verify output and plot	Script executes in validated R environment. Output and plot generated without error.	PASS	None.
OQ-04	Run Python hello-world script	1. Run: jrrun jrhello.py "Hello" 2. Verify output and plot	Script executes in validated Python environment. Output and plot generated without error.	PASS	None.
OQ-05	admin_validate full IQ validation	1. Run admin_validate 2. Verify PASSED result 3. Verify evidence file written	VALIDATION RESULT: PASSED. Evidence file generated.	PASS	Passed after OQ-01 fixes were applied.
OQ-06	Auto-rebuild on requirements change	1. Delete hash file in ~/.renv/[PROJECT_ID]/ 2. Run jrrun jrhello.R 3. Verify auto-rebuild triggered	jrrun detects missing hash, triggers rebuild, then executes script successfully.	PASS	None.
OQ-07	R --add package argument	1. Run admin_install_R --add jsonlite==2.0.0 2. Verify package added 3. Verify R_requirements.txt updated	Package added to local repo and installed. R_requirements.txt updated.	PASS	1 documented limitation (L-01). miniCRAN fetches current CRAN version, not pinned. See Section 8.
OQ-08	Python --add package argument	1. Run admin_install_Python --add requests==2.31.0 2. Verify package added 3. Verify python_requirements.txt updated	Package and all transitive dependencies installed at exact pinned versions.	PASS	pip correctly pinned all transitive dependencies: certifi, charset-normalizer, idna, urllib3.
OQ-09	Integrity check — tampered file	1. Modify a tracked file 2. Run jrrun jrhello.R 3. Restore file and re-run	Step 2: PROJECT INTEGRITY CHECK FAILED. Step 3: runs successfully.	PASS	None.
OQ-10	Bypass	1. Run: Rscript	Both commands	PASS	1 documented

	protection — direct script invocation	R/jrhello.R "Hello" 2. Run: python3 Python/jrhello.py "Hello"	fail without executing in the validated environment.		asymmetry (L-02). Python fails at import rather than with explicit jrrun reference. See Section 8.

6. Performance Qualification (PQ) Results

ID	Script	Procedure	Expected Result	Actual Result
PQ-01	jrhello.R	Run: jrrun jrhello.R "Validation" and capture output	Script runs in validated R environment. Prints message and generates plot. Exit code 0.	PASS — .png file generated showing the string "Validation".
PQ-02	jrhello.py	Run: jrrun jrhello.py "Validation" and capture output	Script runs in validated Python environment. Prints message and generates plot. Exit code 0.	PASS — animated plot generated showing the string "Validation".

7. Findings Summary

The following findings were raised during OQ execution. All findings were resolved prior to OQ completion. No open findings remain.

ID	Test	Severity	Description	Resolution	Status
F-01	OQ-01	Major	renv::restore() installed packages into system library, not the isolated validated library. Environment isolation not functioning.	Replaced renv::restore() with install.packages(lib = lib_path). Added lib.loc = lib_path to verify step.	Closed
F-02	OQ-01	Major	renv::paths$library() resolved relative to working directory, not RENV_PATHS_ROOT. Validation script pointed to wrong library path.	Replaced renv::paths$library() with file.path(renv_root, "library") in admin_generate_validate_R.	Closed

F-03	OQ-01	Minor	admin_R_install.R failed to write renv.lock after clean environment deletion because target directory did not exist.	Added dir.create(dirname(lock_path), recursive=TRUE, showWarnings=FALSE) before writeLines call.	Closed

8. Documented Limitations

The following limitations were identified and accepted. They do not affect the validity of the qualification or the integrity of the validated environment.

ID	Test	Description	Risk Assessment	Action
L-01	OQ-07	miniCRAN fetches current CRAN versions during --rebuild and --add, not pinned versions from R_requirements.txt. R_requirements.txt reflects what is in the repo.	Low. The local repository is the controlled artefact. Rebuilding from CRAN requires re-validation. Normal operation uses the frozen local repo only.	Accept
L-02	OQ-10	Python scripts invoked directly outside jrrun fail at import time rather than displaying an explicit error message referencing the jrrun wrapper.	Low. Scripts cannot execute without validated packages regardless. User still cannot bypass the validated environment.	v1.1

9. Acceptance Criteria

#	Criterion	Met?
1	All IQ steps pass — correct R and Python versions, all packages at pinned versions	Yes
2	All OQ tests pass or have documented accepted findings	Yes
3	All OQ findings are closed or formally accepted as documented limitations	Yes
4	PQ-01 and PQ-02 pass — hello-world scripts execute correctly in	Yes

	validated environments	
5	IQ evidence file attached to Validation Report as Appendix A	Yes
6	Validation Report reviewed and signed off by Validation Manager	Yes

10. Conclusion

The JR Validated Environment v1.0 has successfully completed Installation Qualification, Operational Qualification, and Performance Qualification. All acceptance criteria are met.

Item	Value
IQ result	PASS
OQ result	PASS (3 findings raised, all closed)
PQ result	PASS
Open findings	0
Documented limitations	2 (L-01, L-02 — both accepted)
IQ evidence file	docs/IQ_validation_20260311_205146.txt
Qualification date	2026-03-12
Qualified by	Joep Rous

The JR Validated Environment v1.0 is approved for use in medical device development and regulatory submission workflows.

Appendix A — IQ Evidence File

The IQ evidence file generated by admin_validate is attached below. File path:

docs/IQ_validation_20260311_205146.txt

This file contains the full output of admin_validate, including R and Python version confirmations, all package versions and installation paths, and the VALIDATION RESULT: PASSED verdict.

End of Document — JR-VR-001 v1.0